

Day 14 Observation Guide: Nets & Surface Area

OBSERVER GUIDE

Grade 6 Mathematics · Professor Xavier Honablué, M.Ed. · For visiting guests and observers

Purpose of This Guide

Day 14 is the unit's first move off the flat page. Students unfold boxes into **nets**, discover that every face is a rectangle or triangle whose area they already know how to find, and add those areas to get **surface area**. Nothing new is required mathematically — that is the design. The new work is spatial: keeping track of six faces without losing one.

This guide explains the sequencing decision, maps today to the i-Ready lesson it comes from, and links every material and research source used to build it.

The Design Decision: Cut a Real Box First

Each table group receives an actual cardboard box and scissors. Before any formula appears, groups cut along edges, flatten the box, and trace its net. Students then label each flattened face with its dimensions and discover the three matching pairs themselves. The formula $SA = 2(lw) + 2(lh) + 2(wh)$ is written only after a group has said, in their own words, "the faces come in twos."

We sequence nets here — between triangles (Day 13) and trapezoids (Day 15) — for a specific reason: a triangular prism's net needs both of the area formulas the class has just proven. Surface area becomes the place those formulas get used together, rather than a separate topic at the end of the unit.

Why Surface Area Is Where Units Get Taught Again

Surface area is the moment the difference between square units and cubic units becomes real. Today's exit ticket asks students to explain why wrapping paper is measured in square inches even though a box is three-dimensional. Volume is not taught today; it is deliberately held back so that surface area is not confused with it in the same period.

i-Ready Classroom Mathematics Alignment

Today's lesson is drawn from i-Ready Classroom Mathematics, Grade 6, Unit 1, Lesson 4 ("Use Nets to Find Surface Area"). Lesson 3 (area of polygons, including trapezoids) is taught tomorrow on Day 15 — this order was chosen so the triangle formula from Day 13 is used immediately in the triangular-prism net.

i-Ready content (Grade 6, Unit 1)	Where it appears today	Standard(s)
Lesson 4, Session 1 · Explore — unfolding a prism into a net	Hands-on box cutting; Learn section 1 of the Lesson & Practice packet	6.G.A.4, MP.4
Lesson 4, Session 2 · Develop — surface area of rectangular prisms and cubes	Learn sections 2–3; Worked Examples 1–2	6.G.A.4, MP.7
Lesson 4, Session 2 · Develop — triangular prisms	Learn section 4; Worked Example 3	6.G.A.4, 6.G.A.1
Lesson 4, Session 3 · Refine — real-world surface area and missing dimensions	Worked Example 4; practice items 7–12	6.G.A.4, 6.EE.B.7
Lesson 4 · Math Journal — square units vs. cubic units	Exit ticket prompt	MP.6

Teacher note: session numbers follow the i-Ready Classroom Mathematics Grade 6 Teacher Edition. Confirm exact page numbers against your edition before distributing this guide to observers.

Lesson Flow — 55 Minutes

Min	Segment	What the class is doing	Look for
0–5	Do Now	Two triangle-area problems from Day 13, one with a distractor slanted side.	Retention of $\frac{1}{2}$ and of the perpendicular height.
5–18	Explore (hands-on)	Cut and flatten a real box; trace and label the net; name the matching pairs.	Groups discovering pairs before being told.
18–32	Develop	Worked Examples 1–3: cube, rectangular prism, triangular prism.	Are all six faces (or 2 triangles + 3 rectangles) accounted for?
32–46	Practice	Lesson & Practice packet items 1–12 in pairs, including the missing-height problem.	Organized lists of faces rather than scattered numbers.
46–55	Refine & close	Error analysis (item 6: only three faces counted) and the units journal prompt.	Explanations that say "square units" and why.

Predictable Student Errors — and What We Do About Them

Error	Why it happens	Our move today
Counting only three faces	Students compute lw, lh, and wh once each and stop.	Return to the flattened net and physically point at each twin face.
Losing a face on the net	Nets are drawn without labels, so a face gets skipped.	Require every face to be labeled with its dimensions before any multiplication.
Using cubic units	Surface area is confused with volume.	Exit ticket asks for the unit and the reason; wrapping paper is the anchor image.
Triangular prism: forgetting the two triangles	The rectangles dominate the net visually.	Color-code: triangles in one color, rectangles in another, before computing.

What an Observer Can Look For

- Every group's net is **labeled** before anyone multiplies.
- Students naming faces in pairs ("top and bottom," "front and back," "the two ends").
- Both methods visible: adding six face areas, and the doubled-products formula.
- Answers in **square** units, with a student able to say why.
- The triangular-prism problem using Day 13's $\frac{1}{2}bh$ without re-teaching.

Today's Materials — Direct Links

Every item below is also one click away on the launcher: cpamathlauncher.streamlit.app.

Material	Link
Lesson & Practice packet (12 problems + full solutions)	Day14_Lesson_Packet.pdf
Day 13 Lesson & Practice packet (prerequisite: triangle area)	Day13_Lesson_Packet.pdf
Homework — IXL, Grade 6 skill "Surface area of cubes and rectangular prisms" (90% target)	ixl.com/math/grade-6/surface-area-of-cubes-and-rectangular-prisms
Classroom materials	One empty cardboard box per table group, scissors, tape, rulers, colored pencils.

Standards Addressed

Code	Standard	Where it shows up today
6.G.A.4	Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures; apply these techniques in the context of solving real-world and mathematical problems.	The entire lesson.
6.G.A.1	Find the area of right triangles and other polygons by composing and decomposing.	Triangular-prism net (2 triangles + 3 rectangles).
6.EE.B.7	Solve $px = q$ problems.	Missing-height item 7 and cube-edge item 8.
MP.4	Model with mathematics.	Box cutting; wrapping paper, fish tank, and paint problems.
MP.7	Look for and make use of structure.	Discovering the three congruent pairs of faces.

Research & Further Reading

Every hyperlink below opens the source directly. These are the sources used to build today's lesson.

Source	Why it is here
Common Core State Standards — Grade 6, Geometry (6.G.A.1–4)	The exact wording of the standards this unit teaches, including the "compose and decompose" language our lessons are built on.
IES / What Works Clearinghouse — Improving Mathematical Problem Solving in Grades 4–8 (full PDF)	The federal practice guide behind our daily routine: worked examples with steps, student error analysis, and multiple solution paths.
Curriculum Associates — i-Ready Classroom Mathematics	The publisher page for the curriculum this unit follows (Try–Discuss–Connect instructional routine).
i-Ready Family Center	Free family-facing explanations of each lesson — useful to hand to parents who ask how to help at home.
YouCubed at Stanford — open math tasks	Stanford Graduate School of Education tasks used for our visual and multi-strategy work.
Illustrative Mathematics — Grade 6, Unit 1, Lesson 14: Nets and Surface Area	Open-licensed lesson with printable nets to fold; matches today's hands-on opening.
GeoGebra — Net for a Rectangular Prism	Projectable applet that folds and unfolds a prism; useful for students who cannot picture the flattening.
Khan Academy — Nets of 3-D figures	Short video for absent students and for home review.

If you are observing today, table groups will be cutting boxes for the first fifteen minutes — the room is louder by design. Ask any group to show you their net and name the pairs of faces; that is the check we are making all period.